



UE24EE242B EMF THEORY

~~UNIT - 2~~

Dr. T. S. Chandar

Department of ECE, PESU.

Text Book

1. “Principles of Electromagnetism”,
Matthew N. O. Sadi ku,
4th / 6th Edition, Oxford University
Press, 2007 / 2018.

Reference Books:

2. “Electromagnetic Field Theory
Fundamentals”, Bagh Singh Gur u, Huseyi n
R Hi zi rogl u, Cambr i dge Uni ver si t y
Press, 2nd Edition, 2002.

2. “Engineering Electromagnetism”,
William H Hayt Jr, J. A. Buck, 7th
Edition, Tata McGraw Hill, 2007.

3. “Microwave Devices and Circuits”,
Samuel . Y. Li ao, Thi r d Edi t i on,
Pear son, 2006.

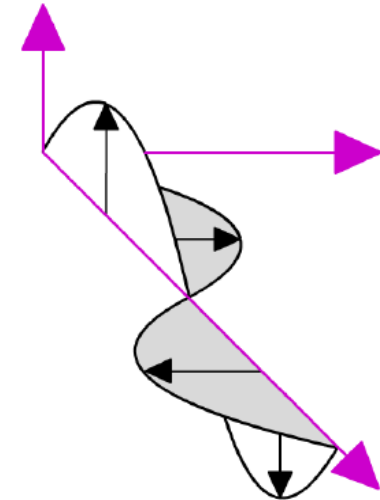
4. “Engineering Electromagnetism



UNIT - 2

Electric Fields in Material Space

Dr. T. S. Chandar
Department of ECE



EMF Theory – Unit – 2 { Text Book Sec. 5.1 – 5.9 & 6.2, 6.4 & 6.5 }



- ❖ Material properties, current and conduct or.
- ❖ Pol ar i sat i on i n Di el ect r i cs.
- ❖ Rel at i on bet ween & i n mat er i al s.
- ❖ El ect r i c Suscept i bi l i t y, Di el ect r i c Constant & t hei r rel at i on.
- ❖ Cont i nui t y Equat i on & Rel axat i on Ti me.
- ❖ El ect r i c Boundary Condi ti ons.
- ❖ Poi sson' s & Laplace' s Equat i ons.
- ❖ Resi stance & Capaci tance

- Conductivity (σ), in mhos / m ($\Omega^{-1} \text{m}^{-1}$) or usually Siemens / m (S/m)
- A material with high conductivity (σ) is referred as **Metal**.
- One with low conductivity (σ) is termed as **Insulator / Dielectric**.
- A material whose conductivity lies somewhere between those metals and insulators is called a **Semiconductor**.
- The conductivity of metals generally increases with decrease in temperature.

EMF THEORY – Unit 1

MATERIAL PROPERTIES

TABLE B.1 Approximate Conductivity* of Some Common Materials at 20°C

Material	Conductivity (siemens/meter)
<i>Conductors</i>	
Silver	6.1×10^7
Copper (standard annealed)	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Tungsten	1.8×10^7
Zinc	1.7×10^7
Brass	1.1×10^7
Iron (pure)	10^7
Lead	5×10^6
Mercury	10^6
Carbon	3×10^4
Water (sea)	4
<i>Semiconductors</i>	
Germanium (pure)	2.2
Silicon (pure)	4.4×10^{-4}
<i>Insulators</i>	
Water (distilled)	10^{-4}
Earth (dry)	10^{-5}
Bakelite	10^{-10}
Paper	10^{-11}
Glass	10^{-12}
Porcelain	10^{-12}
Mica	10^{-15}
Paraffin	10^{-15}
Rubber (hard)	10^{-15}
Quartz (fused)	10^{-17}
Wax	10^{-17}



- The current (in Amperes) through a given area is the electric charge passing through the area per unit time.
- The total current flowing through a surface is
- is the **Current Density**. Depending on how it is produced, there are different kinds of Current Density.
- **Convection Current Density, Conduction Current Density & Displacement Current Density.**
- Currently, we will be discussing the first two

- **Convection Current**, does not involve conductors and hence **does not satisfy Ohm's Law**
- It occurs when the current flows through an insulating medium such as liquid, rarified gas or a vacuum
- The current density at a given point is the current through a unit normal area at that point.
- If there is a flow of charge, at a velocity , through a volume having charge density , then the **Convection Current Density** is given by .

EMF THEORY – Unit - 2

CONVECTION & CONDUCTION CURRENTS



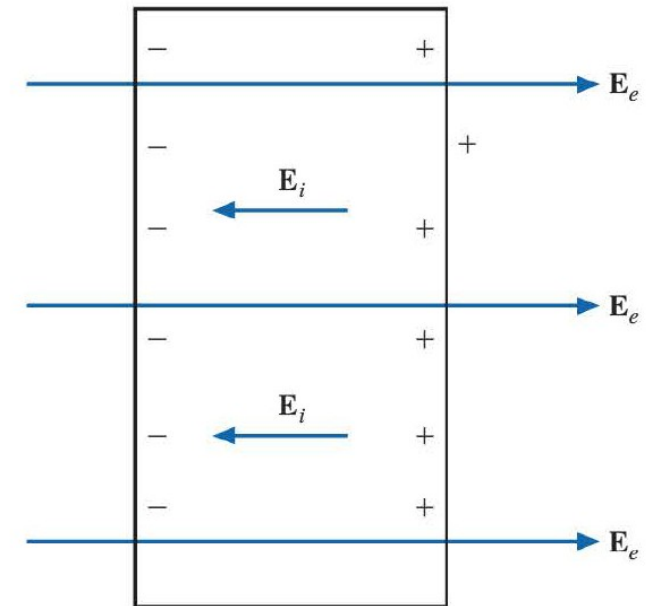
- **Conduction Current**, requires a conductor and hence obeys Ohm's Law
- A conductor is characterised by a large number of free electrons that provide conduction current due to an impressed electric field.
- In this case, the Conduction Current Density is given by;
- This is the point form of Ohm's Law.
- is the conduction current & is in

EMF THEORY – Unit - 2

CONDUCTORS



- A conductor has an abundance of charge that is free to move.
- Case 1: Consider an isolated conductor placed in an field.
- When an external field is applied, the positive free charges are pushed along the same direction as the applied field, while the negative free charges move in the opposite direction. This charge migration takes place very quickly.
- Firstly, these charges accumulate on the surface of the conductor and form an

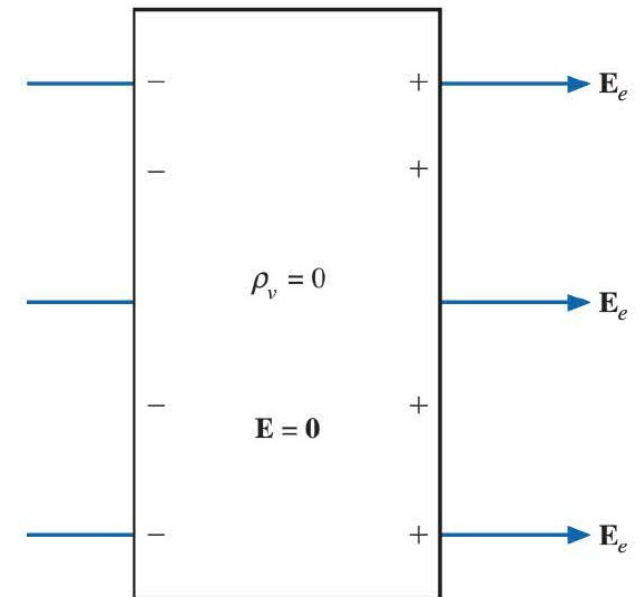


EMF THEORY – Unit - 2

CONDUCTORS



- The induced field cancels the externally applied field.
- A perfect conductor cannot contain an electrostatic field within it.
- A conductor is an equipotential body; the potential is same everywhere in the conductor. This is due to .
- From Gauss' s Law, if , then .
- Summarising, under Static Conditions;
- inside a conductor .

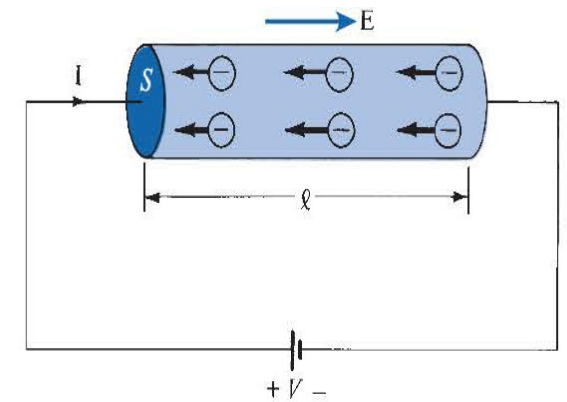


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CONDUCTORS



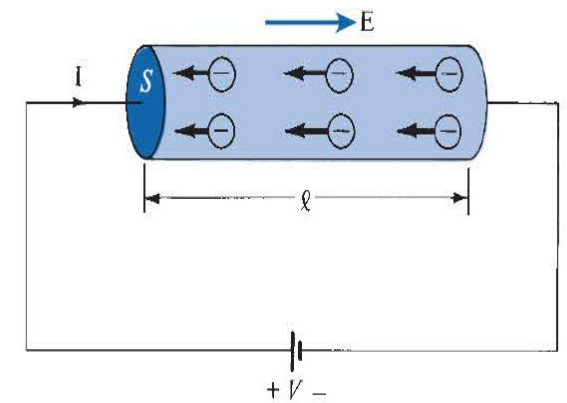
- **Case 2:** Consider a conductor whose ends are maintained at a potential difference.
- In this case inside the conductor. There is no static equilibrium as the conductor is not isolated but wired to a source of electromotive force, which compels the free charges to move and prevents the eventual establishment of electrostatic equilibrium.
- An electrostatic field must exist inside the conductor to sustain the flow of current.



EMF THEORY – Unit - 2

CONDUCTORS

- Suppose the conductor has **uniform** cross section area and is of length l .
- The direction of the electric field produced is the same as the direction of flow of positive charges or current.
- The resistance in such case is given by
- ρ is the **resistivity** of the material.
- If the cross section is **not uniform**, then,

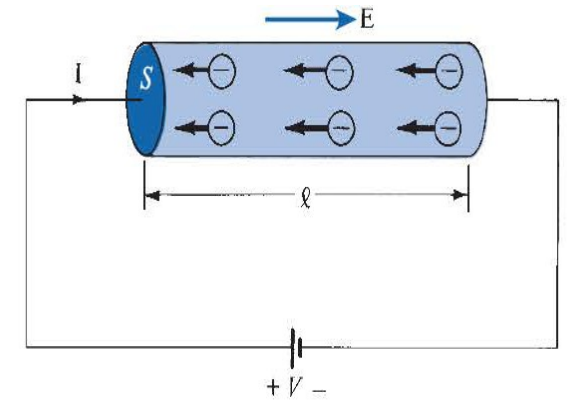


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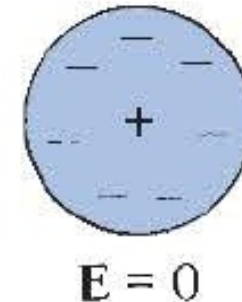
CONDUCTORS



- The negative sign in the numerator of ρ is dropped since ρ is a scalar.
- Other important relations:
- Power (in watts) - **Joule's Law**
- For a conductor with uniform Cross Section, ρ is constant.
- **Numericals.**



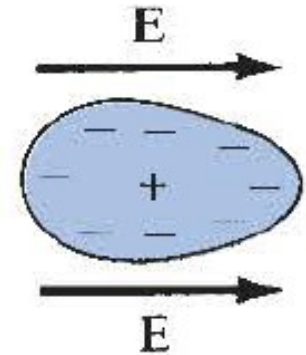
- The main difference between a **conductor** and a **dielectric** lies in the availability of **free** electrons in the outermost atomic shells to conduct current.
- Although the charges in the dielectric are not able to move freely (bounded by finite forces), they can be displaced by an external force.
- Consider an atom of the dielectric consisting of a negative charge (electron cloud) and a positive charge (nucleus) as shown.



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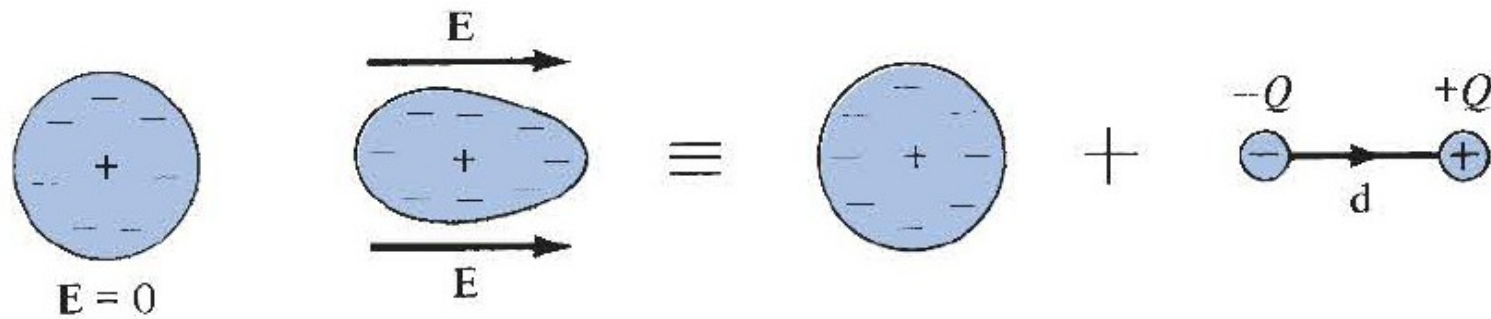
POLARISATION IN DIELECTRICS

- When an electric field is applied, the positive charge is displaced from its equilibrium position in the direction of the force, while the negative charge is displaced in the opposite direction by the force.
- A **di pole** results from the displacement of charges, and the dielectric is said to be **polarised**.
- In the polarised state, the electron cloud is distorted by the applied electric field.



EMF THEORY – Unit - 2

POLARIZATION IN DIELECTRICS

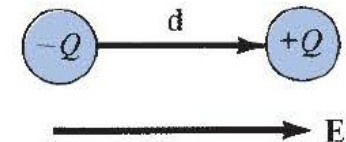
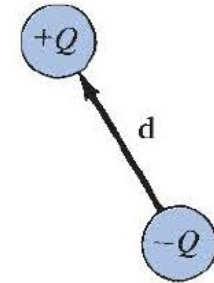


- Where $\vec{r}$ is the distance vector from $+$ to $-$ of the dipole.
- If there are dipoles in a volume of the dielectric, the total dipole moment due to the electric field is
- As a measure of intensity of the polarisation, we now define *polarisation* (in coulombs per sq. m) as the dipole moment per unit volume of the dielectric.
- The major effect of the electric field on a dielectric is the creation of dipole

EMF THEORY – Unit - 2

POLARISATION IN DIELECTRICS

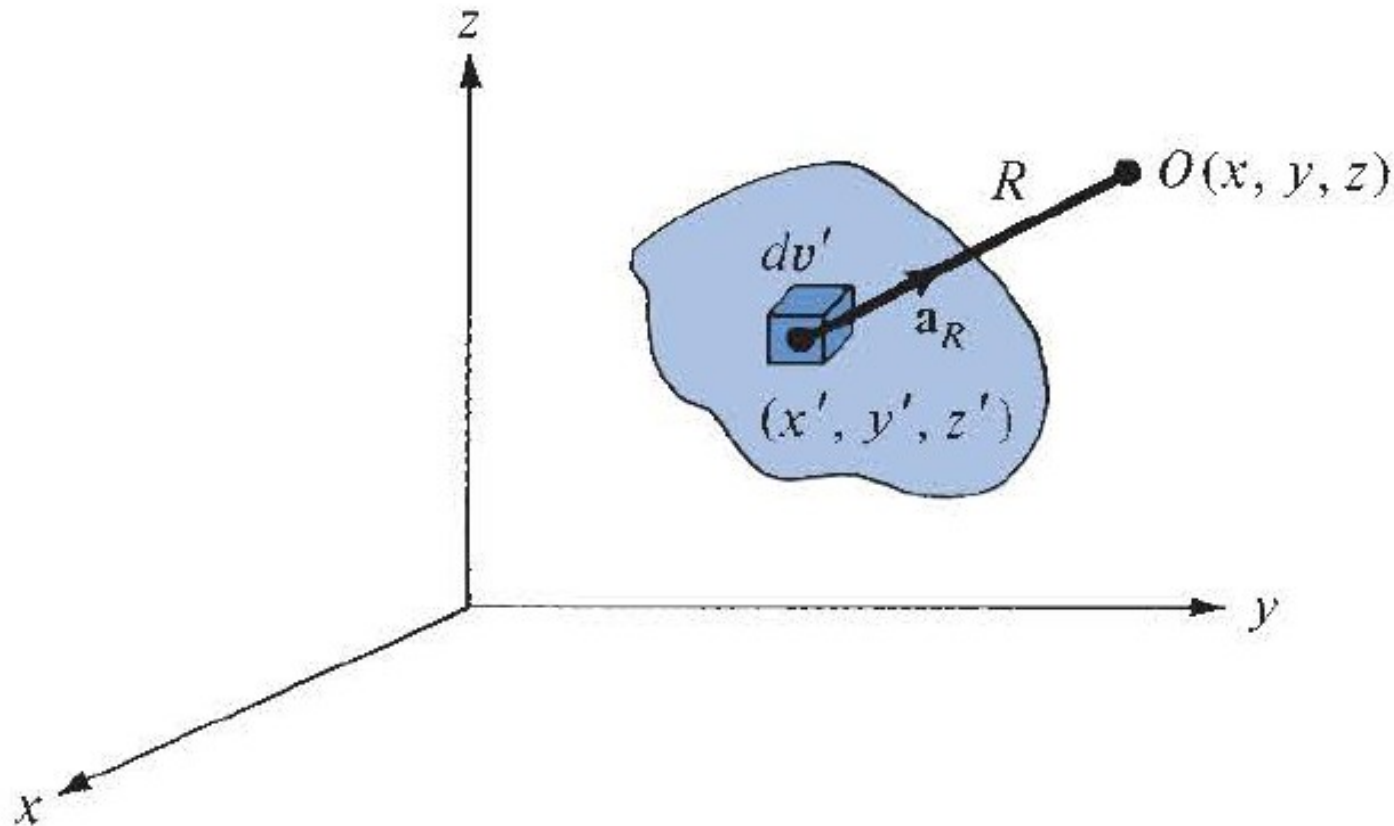
- This type of dielectric is said to be *nonpolar*.
- Examples – Hydrogen, Oxygen, Nitrogen & rare gases.
- Nonpolar dielectric molecules do not possess dipoles until the application of electric field.
- Other types of molecules such as water, sulphur dioxide, HCl etc., have built-in permanent dipoles that are randomly oriented. They are said to be *polar*.



EMF THEORY – Unit - 2

POLARISATION IN DIELECTRICS

- To calculate the field due to a polarised dielectric, consider the following arrangement



- Consider a dielectric material consisting of dipoles with dipole moment per unit volume. Refer the relation;
- The potential at the external point due to the dipole moment is;
- Where r is the distance between the volume at (x, y, z) and the field point (x', y', z')

- We will now transform the expression for $\nabla \cdot \mathbf{D}$ for easy physical understanding. The gradient of $\mathbf{D}$ with respect to the primed co-ordinates is (to be proved later);
- $\nabla' \cdot \mathbf{D} = \rho$; where ∇' is the del operator with respect to (x', y', z')
- Then ; consider
- Applying the vector identity

- Applying the vector identity
- Substituting the above equation in the expression for $\oint \mathbf{E} \cdot d\mathbf{l}$ and integrating over the entire volume of the dielectric, we get;
- Applying Divergence theorem to the first term in RHS gives;

- Comparing the two terms of with the similar expressions for due to surface charge & volume charge distributions,
- ;
- We understand that these two terms denote the potential due to surface and volume charge distributions with densities (upon dropping the primes);
- and
- In other words, the expression for reveals that where polarisation occurs,

- Also, an equivalent surface charge density is formed over the surface of the dielectric.
- and are referred as *bound* (or *polarisation*) *surface* and *volume charge densities*, respectively.
- These are *distinct* from *free* surface and volume charge densities and .
- The total positive bound charge on surface bounding the dielectric is .
- The charge that remains inside is

- If the entire dielectric were electrically **neutral** prior to application of the electric field and if we have **not** added any free charge, the dielectric will continue to remain electrically neutral.
- The total charge, in this case:
- We now consider the case in which the dielectric region **contains** free charge. If ρ_f is volume charge density of the free charge, the total volume charge density is given by;
- (Maxwell's Eqn.)
- Now, ;

- For free space , where .
- To conclude, the net effect of the dielectric on the electric field is to increase inside it by an amount .
- In other words, the application of to the dielectric material causes the flux density to be greater than it would be in free space.
- For some dielectrics, is proportional to the applied electric field . Then , where is known as the *electric susceptibility* of the material.
- *Electric susceptibility* is a measure of how

- The polarisation discussed till now is called *Electronic Polarisation*.
- *Ionic Polarisation* occurs in ionic dielectrics, in which **cations** (positive ions) and **anions** (negative ions) are separated by a certain distance.
- When an external electric field is applied, the two ions are separated further, increasing the dipole moment.
- Ionic polarisation is observed etc.
- *Orientational Polarisation* occurs mostly in liquids & gases.

- With the application of an external field, these dipoles tend to orient or align along the field, resulting in a non-zero polarisation value.
- *Interfacial polarisation* occurs in an insulating system having layers of different dielectrics with different conductivities.
- An external field causes charges to drift from one dielectric to another, and a few charges accumulate at their interfacial boundaries, due to different drift velocities of bound charges on account of different conductivities, creating dipoles.

- We know ϵ_0 and ϵ_r . We now try to obtain another expression for ϵ_r .
- or with ϵ_0
- ϵ_r is called the *permittivity* of the dielectric, ϵ_0 is the permittivity of free space & ϵ_r is the *dielectric constant* or *relative permittivity*.
- The dielectric constant (or relative permittivity) is the ratio of the permittivity of the dielectric to that of free space.

- The concepts discussed till now assumed **ideal** dielectrics. Practically no dielectric is ideal.
- When the **electric field is sufficiently large**, it begins to pull electrons completely out of the molecules, and the **dielectric starts conducting** !
- **Dielectric breakdown** is said to have occurred when a dielectric becomes conducting !
- Dielectric breakdown occurs in all kinds of dielectric materials (solids, liquids & gases) and **depends** on the nature of the material, temperature, humidity and the amount of time that the field is applied.

EMF THEORY – Unit - 2

DIELECTRIC CONSTANT & STRENGTH

- The dielectric strength is the maximum electric field that a dielectric can tolerate or withstand without breaking down.*

TABLE B.2 Approximate Dielectric Constant or Relative Permittivity (ϵ_r) and Strength of Some Common Materials*

Material	Dielectric Constant ϵ_r (Dimensionless)	Dielectric Strength E (V/m)
Barium titanate	1200	7.5×10^6
Water (sea)	80	
Water (distilled)	81	
Nylon	8	
Paper	7	12×10^6
Glass	5–10	35×10^6
Mica	6	70×10^6
Porcelain	6	
Bakelite	5	20×10^6
Quartz (fused)	5	30×10^6
Rubber (hard)	3.1	25×10^6
Wood	2.5–8.0	
Polystyrene	2.55	
Polypropylene	2.25	
Paraffin	2.2	30×10^6
Petroleum oil	2.1	12×10^6
Air (1 atm)	1	3×10^6

EMF THEORY – Unit - 2

LINEAR, ISOTROPIC & HOMOGENEOUS

DIELECTRICS

- A dielectric material (in which ϵ applies) is linear if ϵ does not change with the applied field, homogeneous if ϵ does not change from point to point, and isotropic if ϵ does not change with direction.
- A conducting material (in which σ applies) is linear if σ does not change with the applied field, homogeneous if σ does not change from point to point, and isotropic if σ does not change with direction.
- Numericals.



EMF Theory – Unit - 2

CONTINUITY EQUATION



- **Principle of Charge Conservation** – The time rate of ‘decrease’ of charge within a given volume must be equal to the ‘net outward’ current flow through the surface of the volume.
- Then ; – current density.
- Involving Divergence Theorem we can write;
- Also,
- Using these two relations, we get :

EMF Theory – Unit - 2

CONTINUITY EQUATION



- **Continuity Equation or Continuity of Current Equation.**
- Continuity Equation essentially states that there can be no accumulation of charges at any point when currents are flowing.
- For steady (DC) currents, and hence , indicating the total charge leaving a volume is the same as the total charge entering it.

EMF Theory – Unit - 2

RELAXATION TIME



- Using the following (Ohm's Law) and (Gauss's Law - and), we can write the Continuity Equation as; (- Conductivity ρ_{hos}/m or S/m)
- . Solving this equation leads to
- or
- , where - Relaxation time in seconds.

EMF Theory – Unit - 2

RELAXATION TIME



- ρ_v is the initial volume charge density at $t = 0$.
- The equation $\frac{d\rho_v}{dt} = -\frac{\rho_v}{\tau}$ shows that the introduction of charge at some interior part of the material results in the decay of the volume charge density.
- Associated with the decay is the charge movement from the interior point at which it was introduced, to the surface of the material.
- **Relaxation time** or **Rearrangement Time** is the time it takes a charge placed in the

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



- Till now we have considered the field in homogeneous medium
- If the field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the media are called '**Boundary Conditions**'.
- These conditions are helpful in determining the field on one side of the boundary if the field on the other side is known.
- Subsequent discussions would be for the following interfaces separating;

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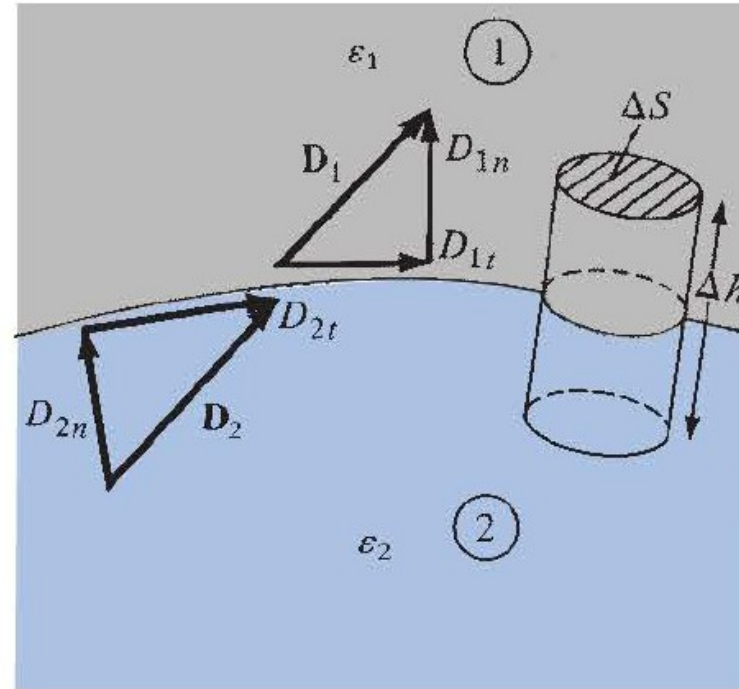
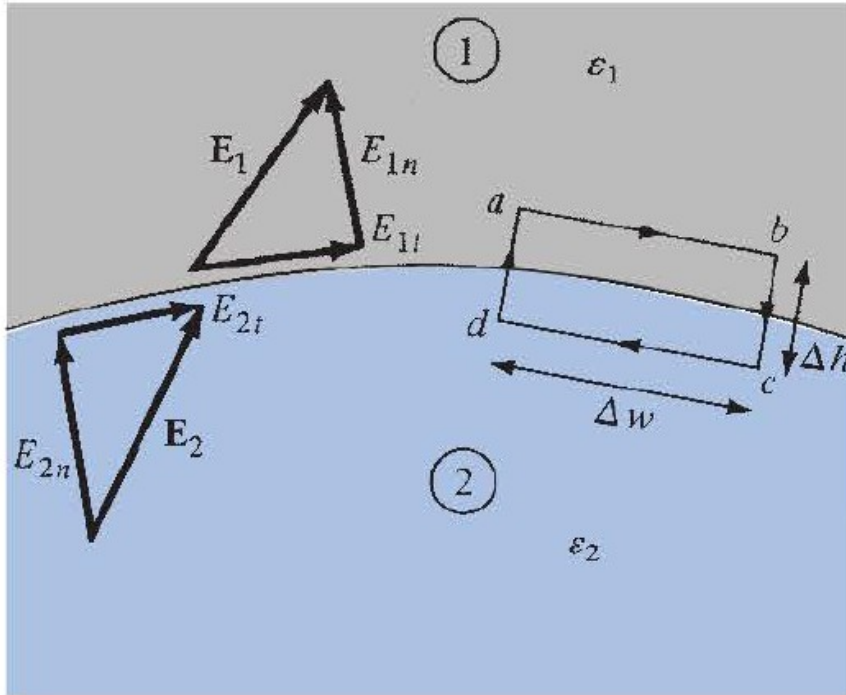
ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- To determine the boundary conditions, we shall use the following Maxwell's Eqns. ;
-
- The electric field intensity is decomposed into two orthogonal components; , its **tangential** and **normal** components.
- Similarly, the electric flux density can also be decomposed to its tangential and normal components.



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ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

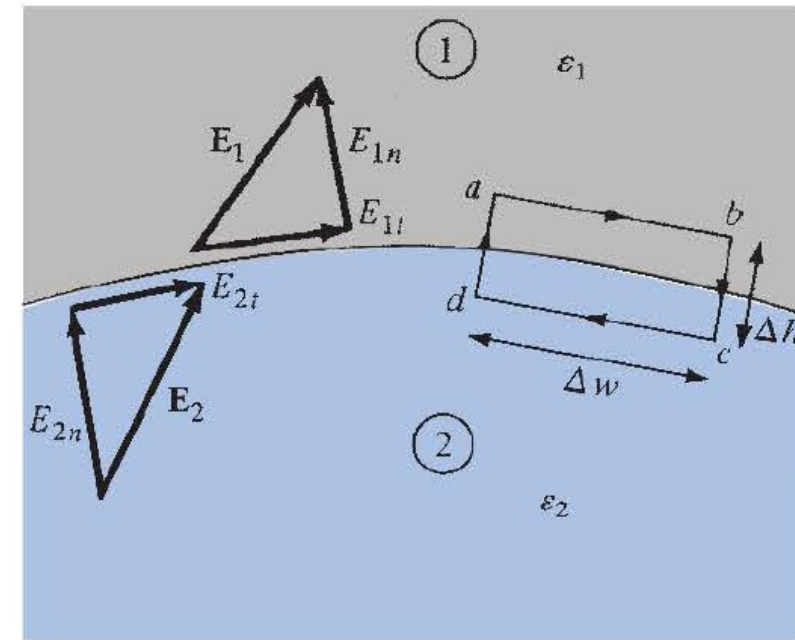


**Dielectric – Dielectric Boundary
Conditions**

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- Consider the field existing in a region that consists of two different dielectrics characterised by ϵ_1 and ϵ_2 . The fields E_1 and E_2 in media 1 and 2 can be decomposed into;
 - E_{1n} and E_{2n} normal to the boundary
 - E_{1t} and E_{2t} tangential to the boundary
- and
- While applying the relation to the closed path, assuming that the path is very small w.r.t spatial variations, we have,

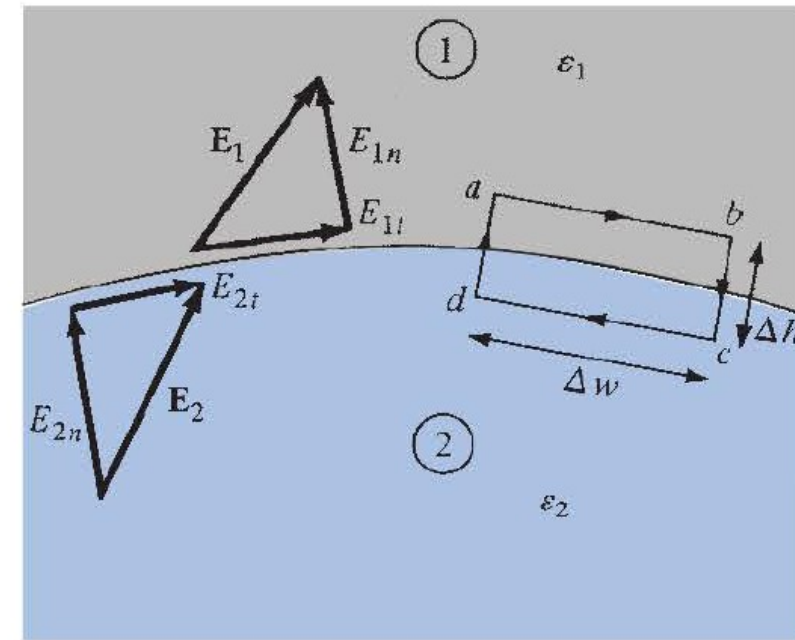


(a)

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- The signs are obtained from the direction of travel along , and the direction of the component vectors.
- Also, and . The terms cancel out resulting in;
- or
- , meaning the **tangential components** of the field are **same** on both sides of the boundary or interface.
- In other words, undergoes no change and said to be '**continuous**' across the



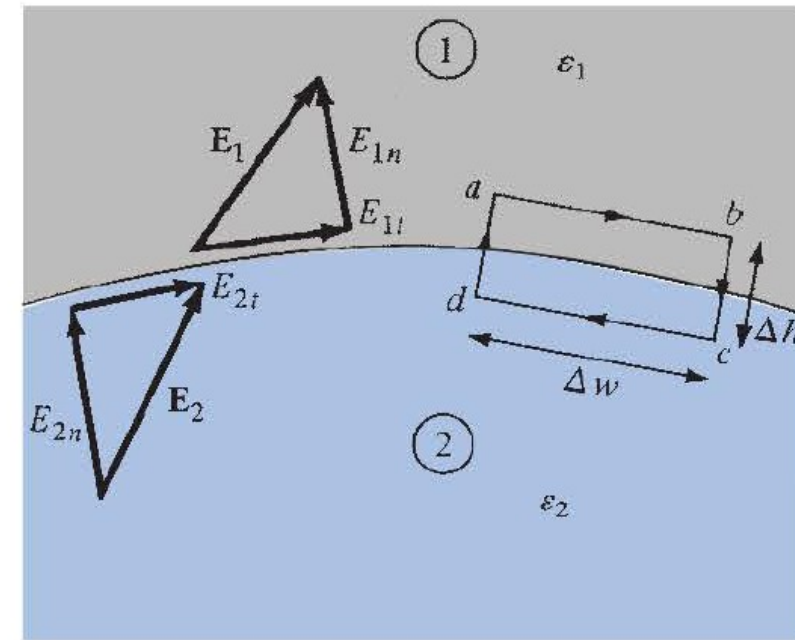
(a)

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- Since , the relation can be expressed as;

-
- Therefore, undergoes some change across the boundary and said to be **discontinuous** across the interface.

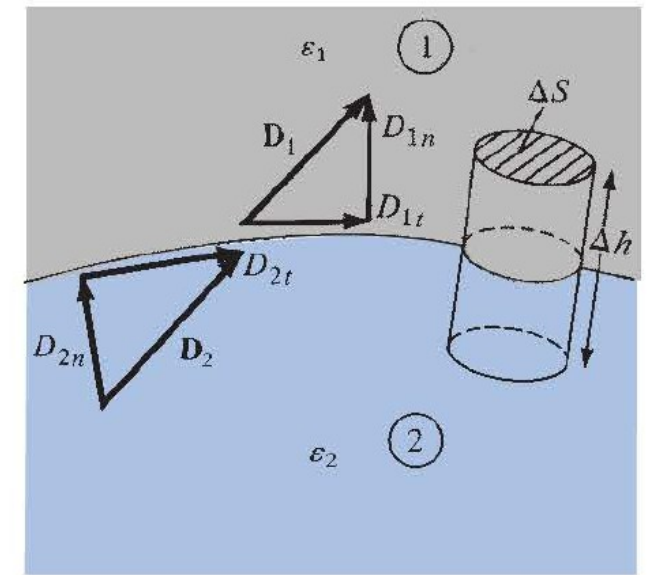


(a)

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- Similarly, when we apply the relation to the **Cylindrical Gaussian Surface**, as shown, the contribution due to the sides vanishes !
- Allowing , gives,
- (-ve for the bottom)
- Hence , Where is the free charge density deliberately placed at the boundary or interface.
- It should be noted that the relation is based on the assumption that is

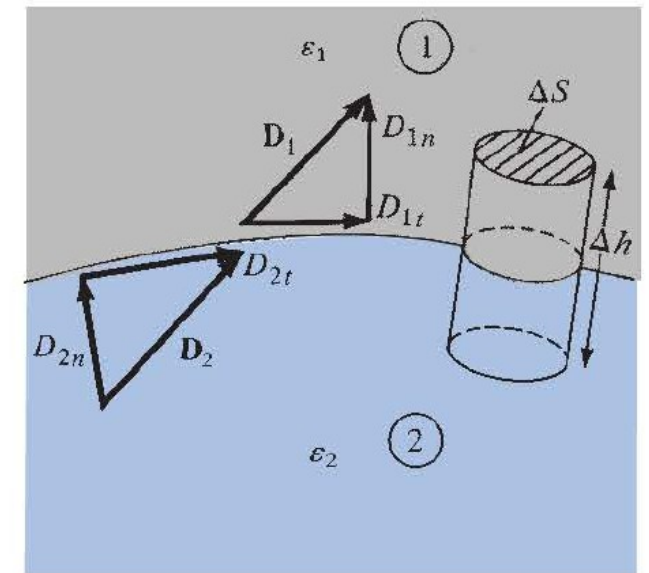


(b)

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- If no free charges exist at the boundary (i.e., not deliberately placed!), then .
- We then have the following relation;
- with .
- Thus, the **normal component** of $\mathbf{D}$ does not undergo a change, **in this case**, across the boundary.
- Alternatively, the normal component of $\mathbf{D}$ is '**continuous**' across the boundary with

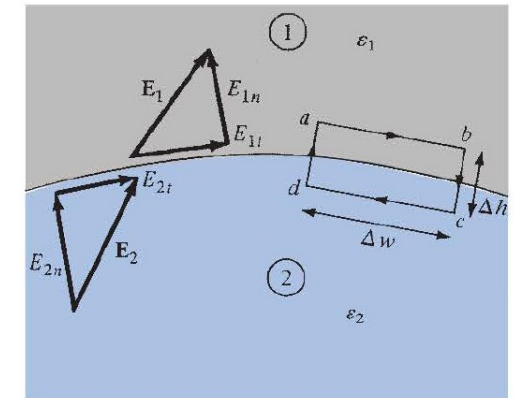


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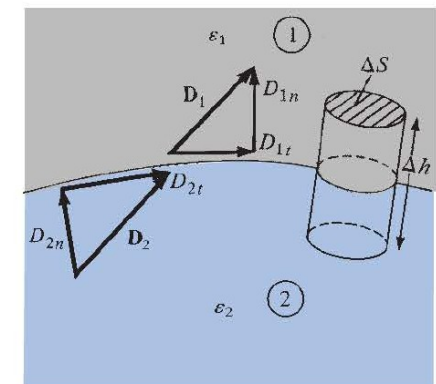
EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- Since $\epsilon_1 \neq \epsilon_2$, we have $E_{1n} \neq E_{2n}$, meaning the normal component of E is discontinuous across the boundary.
- Summarising the boundary conditions that need to be satisfied by an electric field at the boundary separating two dielectrics;
- $E_{1t} = E_{2t}$
- $D_{1n} = D_{2n}$, with



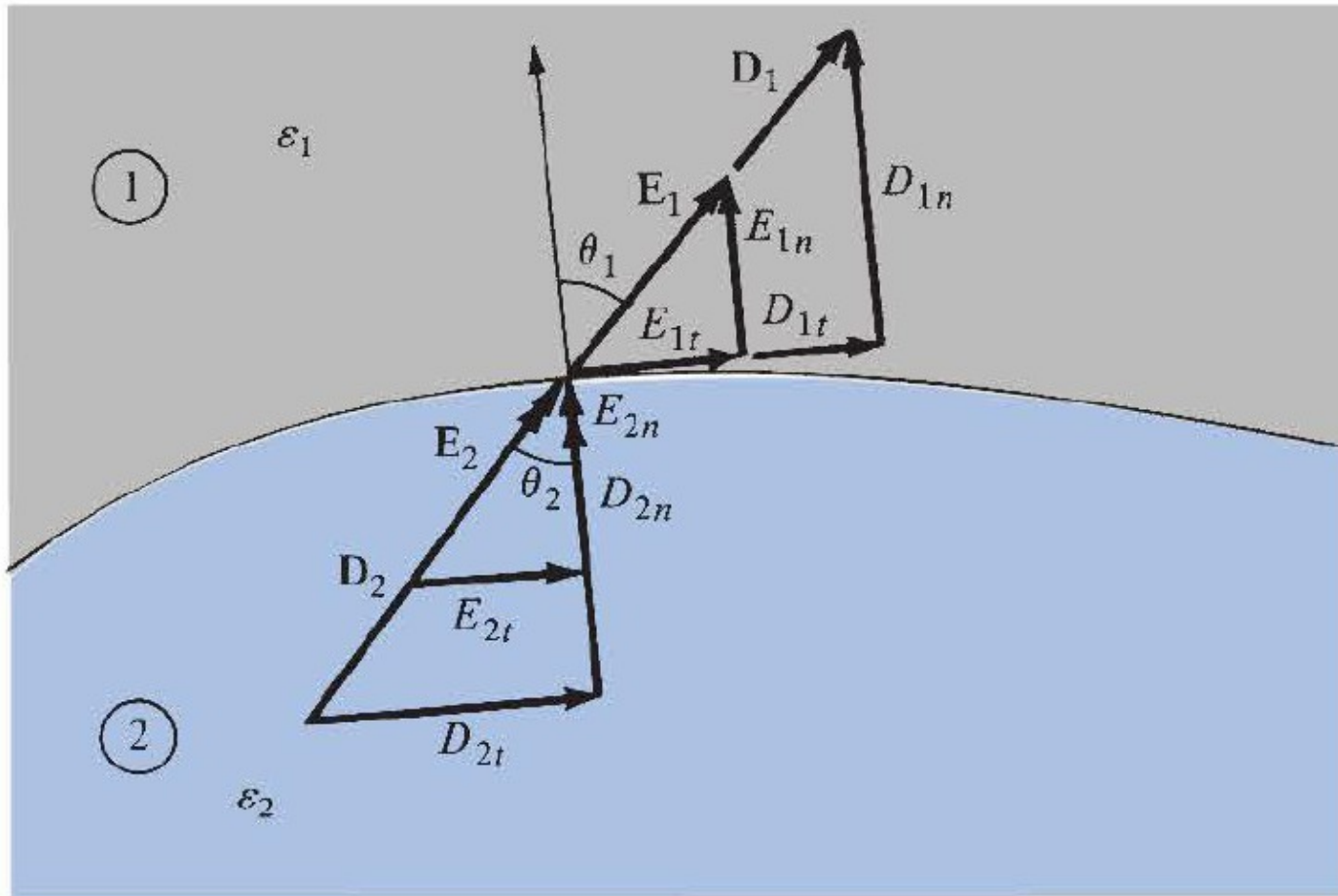
(a)



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EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



Refraction in Dielectric – Dielectric Boundary

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



- Consider or and or making angles and with the 'normal' to the interface as shown.

- Using the relation , we have

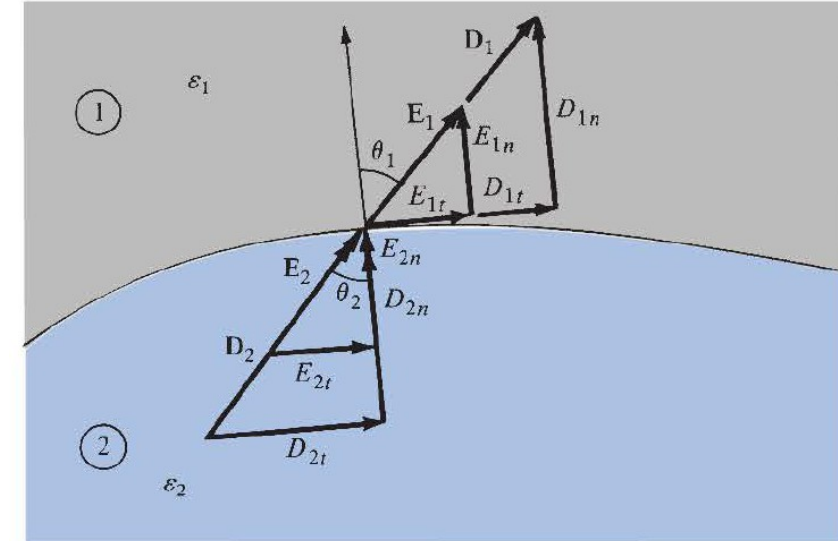
- or

- (1)

- Similarly by applying

- , we have

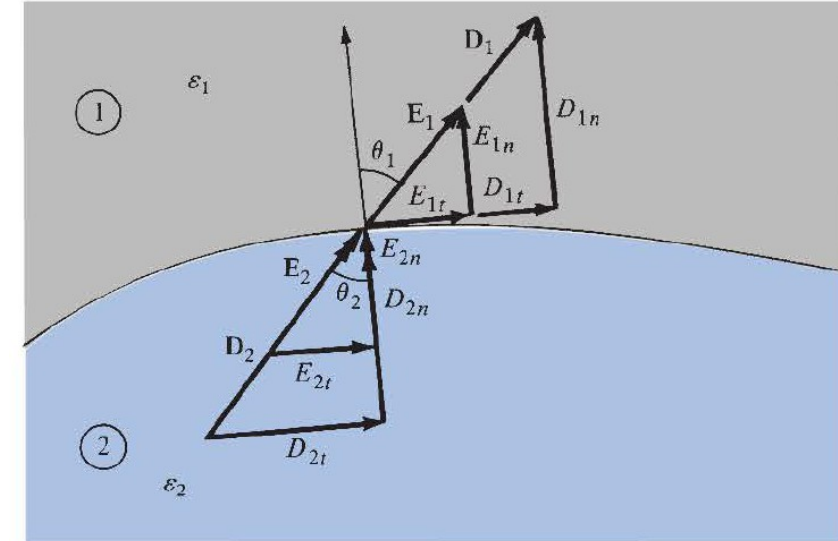
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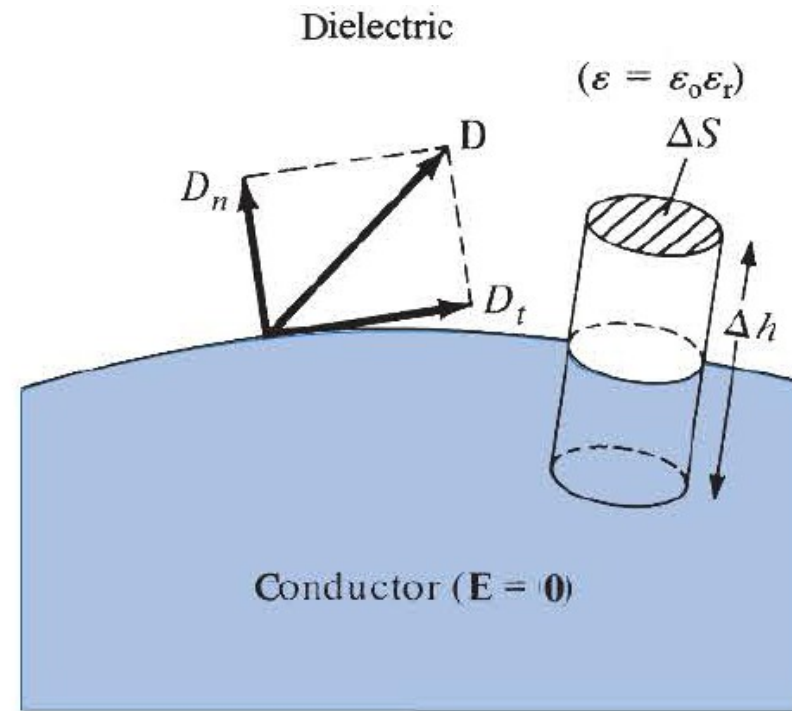
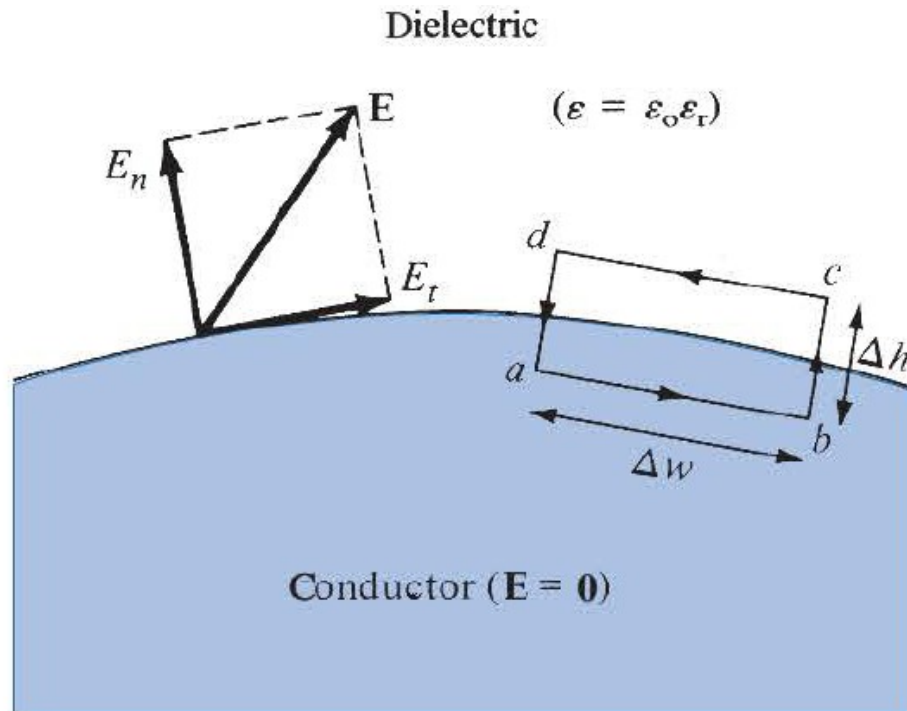
ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- Using both equations, we have;
-
- Since $\epsilon_1 \neq \epsilon_2$ and $\sigma = 0$, we have;
-
- This is the '**law of refraction**' of the electric field at the boundary, free of charge (with $\sigma = 0$).
- In general, an interface between 2 dielectrics produces bending of electric flux lines.



EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



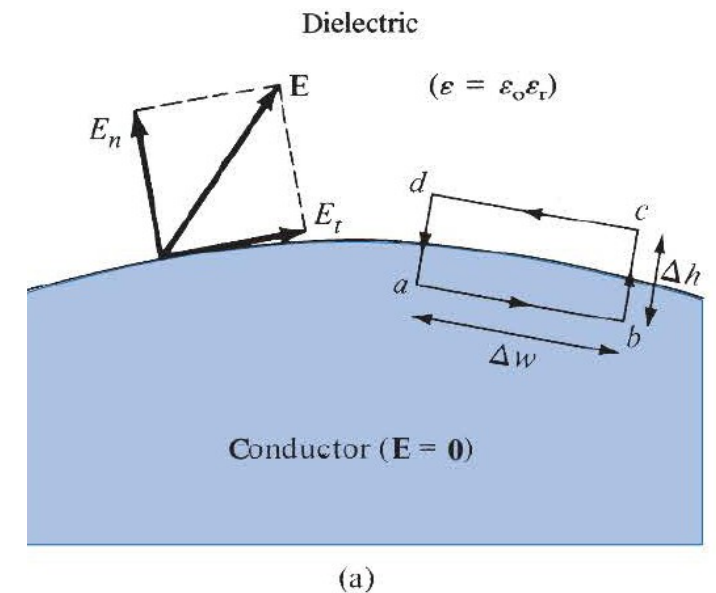
Conductor – Dielectric Boundary
Conditions

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



- The conductor is assumed to be perfect ($\sigma \rightarrow \infty$).
- To determine the boundary conditions, we follow the same procedure, except that inside the conductor.
- While applying the relation to the closed path gives;

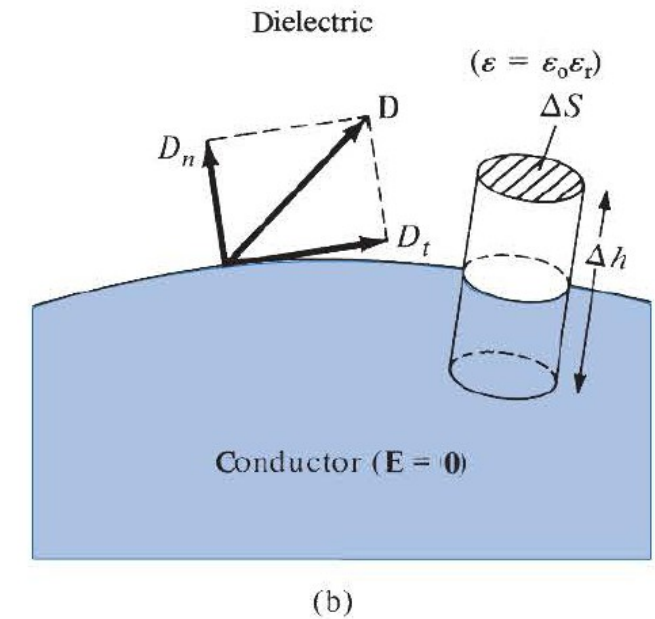


- As ,

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ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

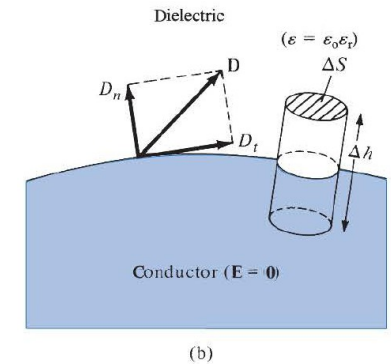
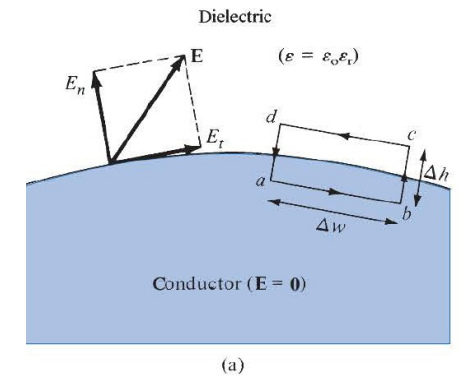
- Similarly, using the relation to the cylindrical Gaussian surface as shown, and letting $\Delta h \rightarrow 0$, we have,
- Since inside the conductor, we have,
- Therefore, under static conditions, following can be inferred / concluded for a ‘**perfect**’ conductor;



EMF Theory – Unit - 2

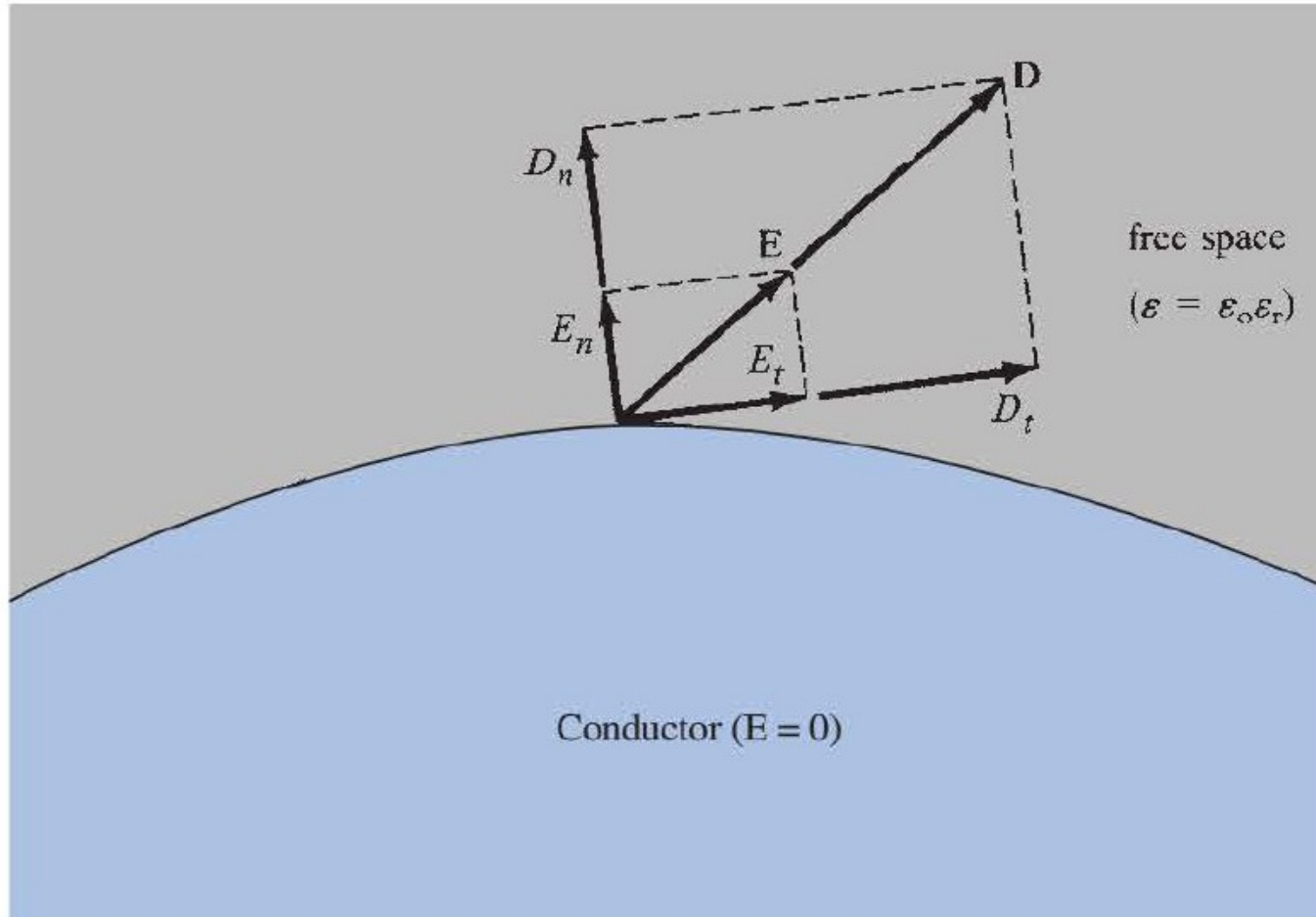
ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

- No electric field may exist within the conductor;
- Since , there can be no potential difference between any two points in the conductor or a conductor is an equipotential body.
- An electric field must be external to the conductor and must be normal to its surface.
- and



EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



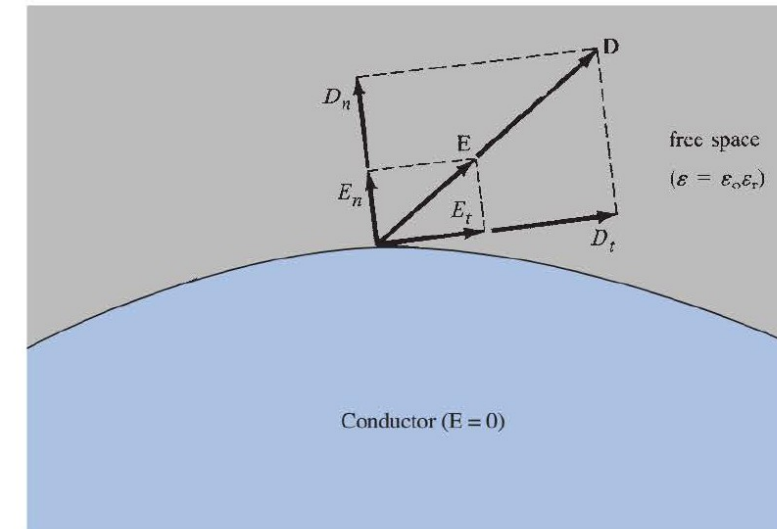
**Conductor – Free Space
Boundary Conditions**

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



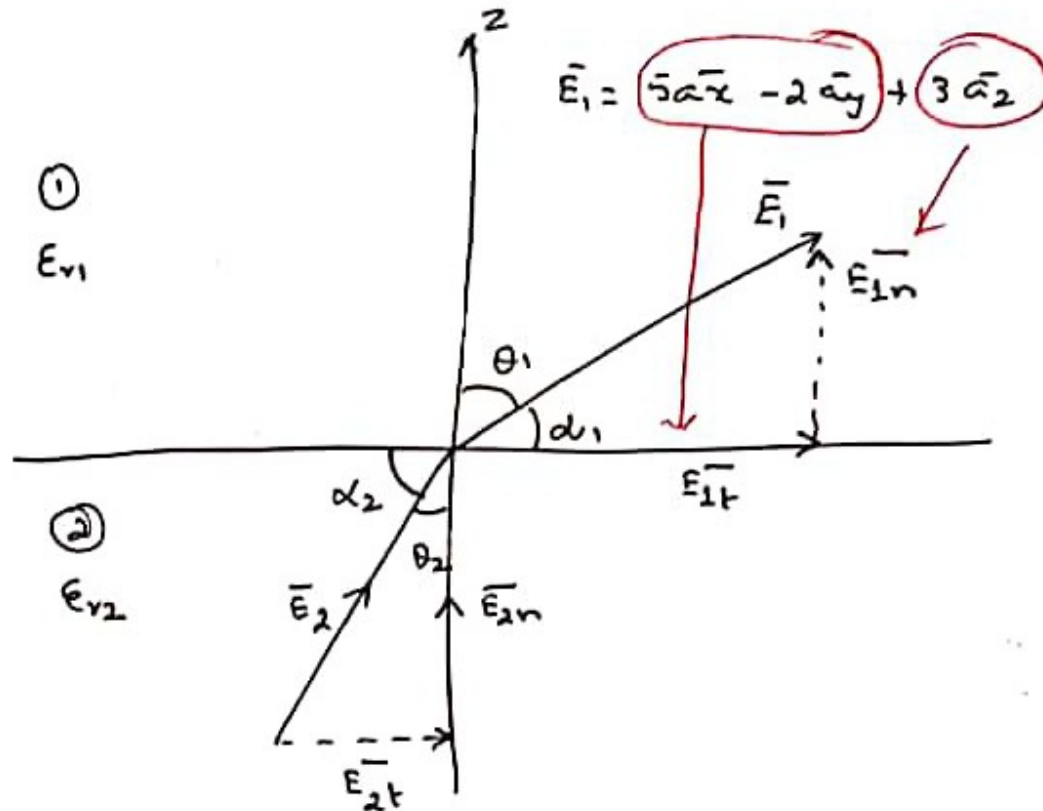
- Boundary conditions can be obtained for the relations; and
- Replacing for free space, we get
- and
- An electric field must be external to the conductor and must be normal to its surface.
- The above relation implies that the field should approach the conducting surface normally.



EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

Two extensive homogeneous, isotropic dielectrics (23) meet on plane $z=0$. For $z>0$; $\epsilon_{r1}=4$, & for $z<0$ $\epsilon_{r2}=3$. A uniform electric field $\vec{E}_1 = 5\vec{a}_x - 2\vec{a}_y + 3\vec{a}_z$ kV/m exists for $z \geq 0$.



EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



(a) Find $\vec{E}_2$ for $z \leq 0$

Since $\vec{a}_2$ (unit vector in z -direction) is normal to the boundary plane, we can obtain the normal component as

$$E_{1n} = \vec{E}_1 \cdot \vec{a}_n = (5\vec{a}_x - 2\vec{a}_y + 3\vec{a}_z) \cdot (\vec{a}_z) = 3.$$

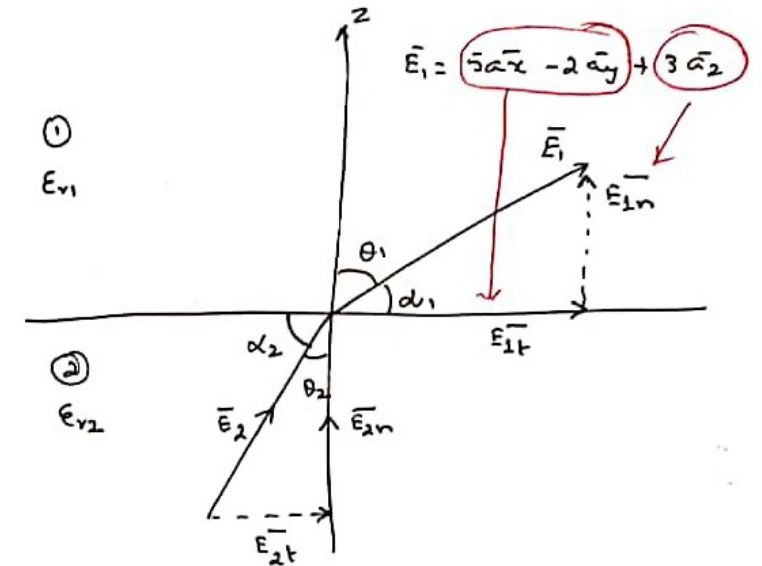
$$\therefore \vec{E}_{1n} = 3\vec{a}_z \text{ (in the } +ve \text{ } z \text{ direction)}$$

Similarly $\boxed{\vec{E}_{2n} = (E_2 \cdot \vec{a}_z) \vec{a}_z}$

$$\text{Now } \vec{E} = \vec{E}_t + \vec{E}_n \Rightarrow E_{1t} = \vec{E}_1 - \vec{E}_{1n} = \underbrace{5\vec{a}_x - 2\vec{a}_y + 3\vec{a}_z}_{5\vec{a}_x - 2\vec{a}_y} - 3\vec{a}_z$$

Now using the boundary conditions of D-D, we have

$$\vec{E}_{1t} = \vec{E}_{2t} \text{ Hence } \underline{\underline{\vec{E}_{1t} = \vec{E}_{2t} = 5\vec{a}_x - 2\vec{a}_y}}$$



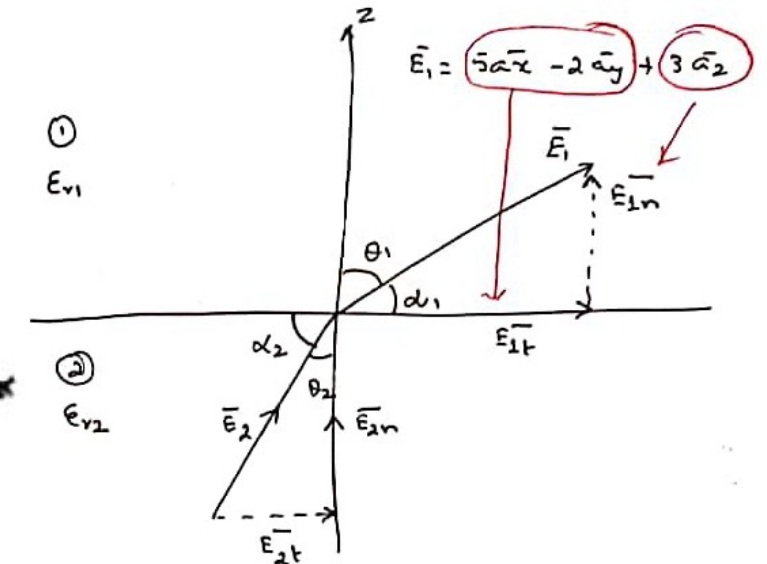
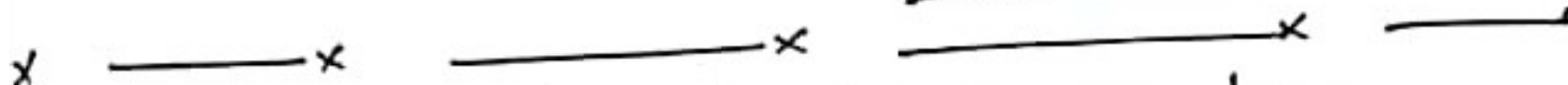
EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

Similarly $\underline{\underline{D_{1n} = D_{2n}}}$ or $\underline{\underline{E_{v1} \epsilon_{1n} = E_{v2} \epsilon_{2n}}}$ (24)

or $\epsilon_{2n} = \frac{\epsilon_{v1}}{\epsilon_{v2}} \epsilon_{1n} = \frac{4}{3} (3\bar{a}_2) = \underline{\underline{4\bar{a}_2}}$

Thus $\underline{\underline{\vec{E}_2 = \vec{E}_{2t} + \vec{E}_{2n} = 5\bar{a}_x - 2\bar{a}_y + 4\bar{a}_2 \text{ kV/m}}}$



EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



(b) The angles $\vec{E}_1$ & $\vec{E}_2$ make with the interface.
from the diagram $\alpha_1 = 90^\circ - \theta_1$ & $\alpha_2 = 90^\circ - \theta_2$

method-1 Now $|\vec{E}_{1n}| \propto E_{1n} = 3$; $|\vec{E}_{1t}| = E_{1t} = \sqrt{5^2 + (-2)^2} = \underline{\underline{\sqrt{29}}}$

$$\text{Hence } \tan \theta_1 = \frac{|\vec{E}_{1t}|}{|\vec{E}_{1n}|} = \frac{E_{1t}}{E_{1n}} = \frac{\sqrt{29}}{3}$$

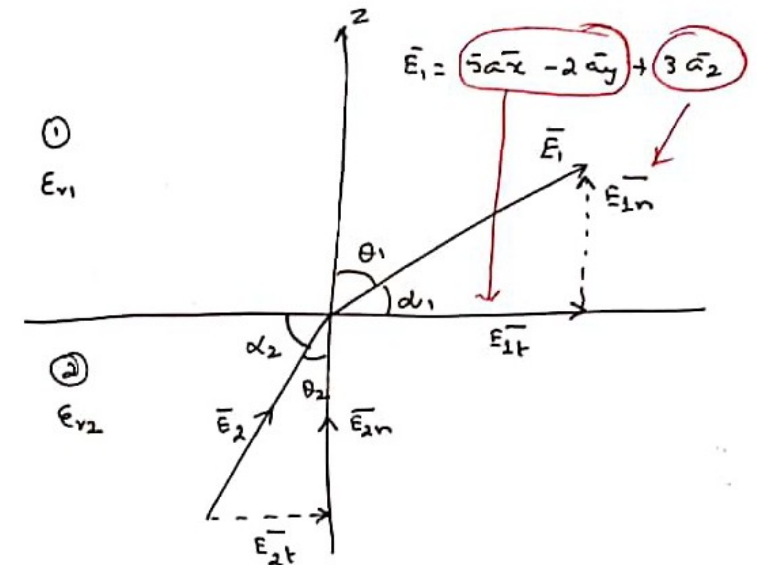
$$\text{or } \theta_1 = \tan^{-1}\left(\frac{\sqrt{29}}{3}\right) = \underline{\underline{60.8784^\circ}}$$

$$\therefore \alpha_1 = 90^\circ - 60.8784 = \underline{\underline{29.1216^\circ}}$$

method-2 $\vec{E}_1 \cdot \vec{a}_n = |\vec{E}_1| \cdot |\vec{a}_n| \cos \theta_1$

$$\text{or } \cos \theta_1 = \frac{\vec{E}_1 \cdot \vec{a}_n}{|\vec{E}_1| (1)} \quad \text{since } \vec{a}_n = \vec{a}_2 = \frac{3}{\sqrt{5^2 + (-2)^2 + 3^2}}$$

$$\cos \theta_1 = \frac{3}{\sqrt{38}} = \underline{\underline{60.8784^\circ}} \Rightarrow \alpha_1 = 90^\circ - 60.8784 = \underline{\underline{29.1216^\circ}}$$



EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

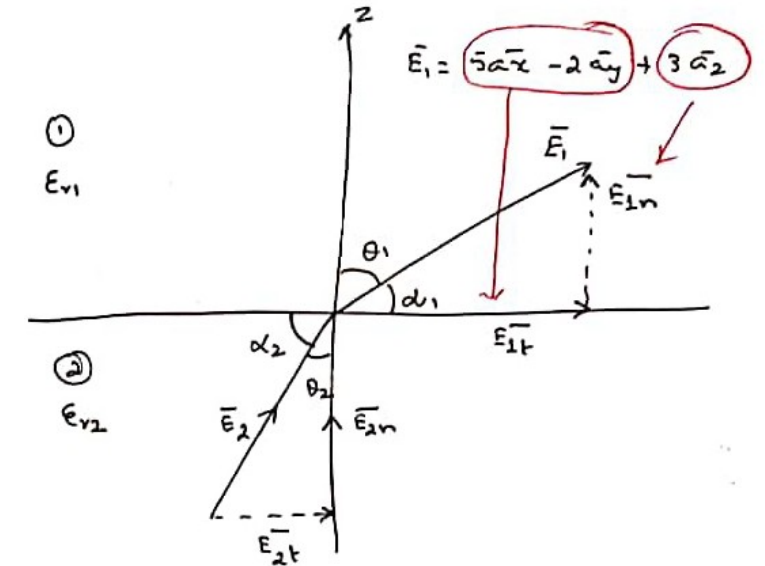


Similarly $|\vec{E}_{2n}| = |4\vec{a}_2| = 4$; $E_{2t} = E_{1t} = \sqrt{29}$

$$\therefore \tan \theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{29}}{4} = 1.346 \Rightarrow \theta_2 = \tan^{-1}(1.346) = \underline{\underline{53.3957^\circ}}$$

Hence $\alpha_2 = 90^\circ - \theta_2 = \underline{\underline{36.6043^\circ}}$

Check $\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}} \Rightarrow \underline{\underline{1.333}}$



EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS

(c) The energy densities in both dielectrics. (25)

Recall $w_E = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon \mathbf{E}^2$ $E = |\vec{E}|$: $\epsilon = \epsilon_0 \epsilon_r$

Then
 $w_{E1} = \frac{1}{2} \epsilon_1 |\vec{E}_1|^2 = \frac{1}{2} \times \frac{4 \times 10^{-9}}{36\pi} \times (25+4+9) \times 10^6$

$\underbrace{\epsilon_1 = \epsilon_0 \epsilon_{r1}} \quad \underbrace{|\vec{E}_1|^2}$

$= 672 \mu\text{J}/\text{m}^3$

Similarly

$$w_{E2} = \frac{1}{2} \epsilon_2 |\vec{E}_2|^2 = \frac{1}{2} \times \frac{3 \times 10^{-9}}{36\pi} \times (25+4+16) \times 10^6$$

$= 597 \mu\text{J}/\text{m}^3$

EMF Theory – Unit - 2

ELECTRIC FIELDS IN MATERIAL SPACE – BOUNDARY CONDITIONS



d) The energy within a cube of side 2m centred at (3, 4, -5).

At the centre (3, 4, -5), of the cube of side 2m, $z = -5 < 0$
i.e. the cube is in Region ② with $2 \leq x \leq 4$, $3 \leq y \leq 5$
4 $-6 \leq z \leq -4$. Therefore

$$W_E = \int W_{E2} dv = \int_{x=2}^4 \int_{y=3}^5 \int_{z=-6}^{-4} W_{E2} dz dy dx.$$
$$= W_{E2} (2)(2)(2) = 597 \times 8 = \underline{\underline{4.776 \text{ mJ}}}$$

EMF Theory – Unit - 2

POISSON'S AND LAPLACE'S EQUATION

- The Electric Field can be determined either using Coulomb's law or Gauss's law, when the charge distribution is known.
- It can also be found out using when the potential is known throughout the region.
- In most practical conditions, such distributions are rarely known / given.
- If the charge or potential is known at 'some' boundaries, we can still compute or using Poisson's or Laplace's Equations



EMF Theory – Unit - 2

POISSON'S AND LAPLACE'S EQUATION



- For a linear, isotropic material medium
- ;
- Using the above relations, we have the following;
 - - Poisson's Eqn. for inhomogeneous medium
 - - Poisson's Eqn. for homogeneous medium
- If (charge free region), then
 - - Laplace's Eqn. for homogeneous medium
 - - Laplace's Eqn. for inhomogeneous medium

EMF Theory – Unit - 2

POISSON'S AND LAPLACE'S EQUATION



- The Laplace's Equation in different co-ordinate systems are;

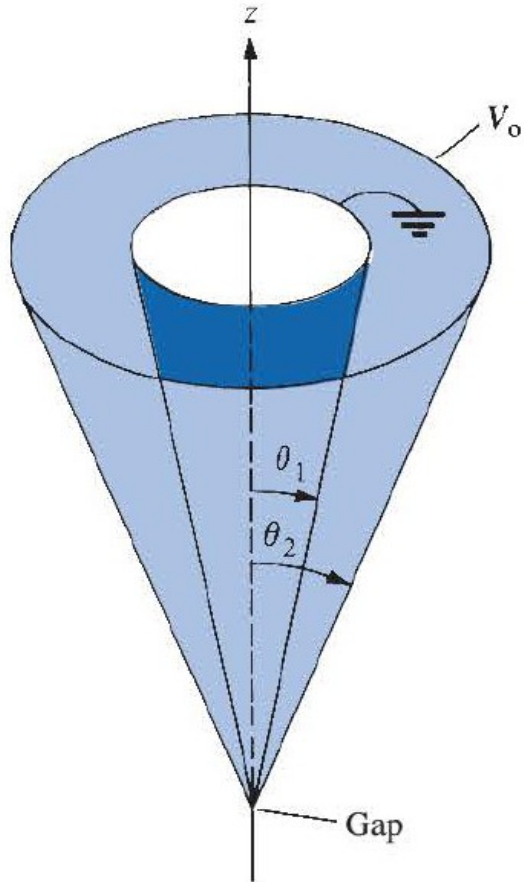
General procedure for solving Poisson's or Laplace's Equation;



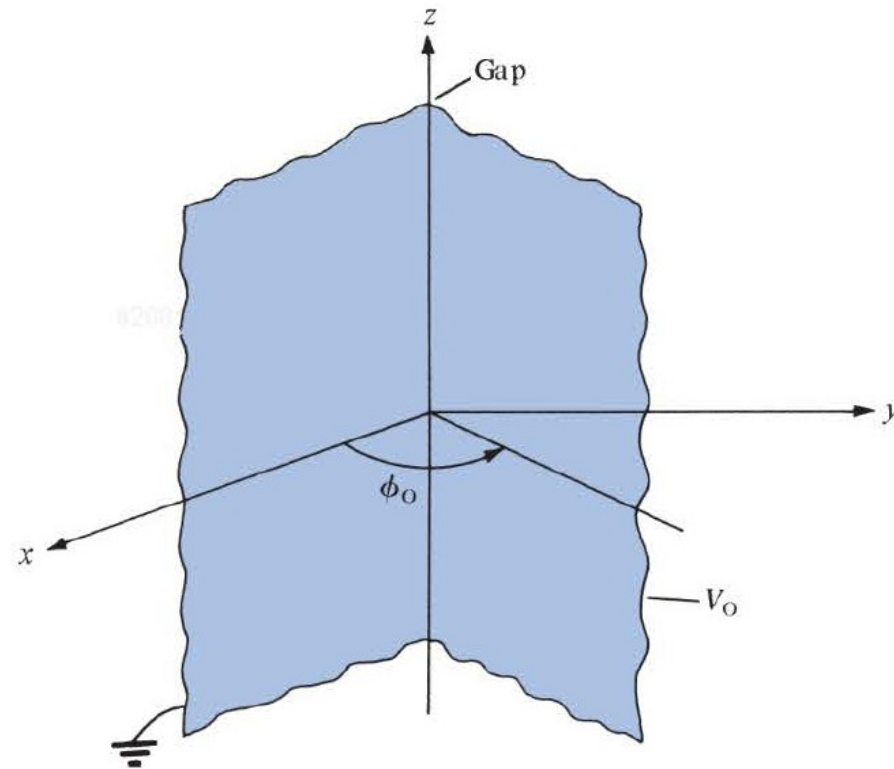
- **1.** Solve Laplace's (if) or Poisson's (if) equation using either (a) direct integration when is a function of one variable or (b) separation of variables if is function of more than one variable.
- The solution at this point is not unique but is expressed in terms of the unknown integration constants that need to be determined.
- **2.** Apply the boundary conditions to determine the unique solution for .

EMF Theory – Unit - 2

POISSON'S & LAPLACE'S EQUATIONS



Ex.
2



Ex.
1

- The resistance of a conductor having **uniform** cross section is given by, $R = \frac{\rho l}{A}$. While the resistance of a conductor having **non-uniform** cross section is given by, $R = \frac{\rho l}{A}$.
- The problem of evaluating the resistance for a conductor having **non-uniform cross section**, can also be treated as 'boundary-value' problem
- Procedure is as follows;
 - **1.** Choose a suitable coordinate system
 - **2.** Assume V as the potential difference between the conductor terminals.
 - **3.** Solve the Laplace's Eqn. to obtain V .

EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



- Then determine $\mathcal{E}$ from $\mathcal{E} = \oint \vec{E} \cdot d\vec{l}$ and find $\mathcal{E}$ from $\mathcal{E} = -\frac{d\phi}{dt}$.
- **4.** Finally, obtain $\mathcal{E}$ as $\mathcal{E} = \frac{d\phi}{dt}$.
- **Alternatively,** it is possible to assume the current i , find the corresponding potential difference V , and determine $\mathcal{E}$ as $\mathcal{E} = \frac{dV}{dt}$.

EMF Theory – Unit - 2

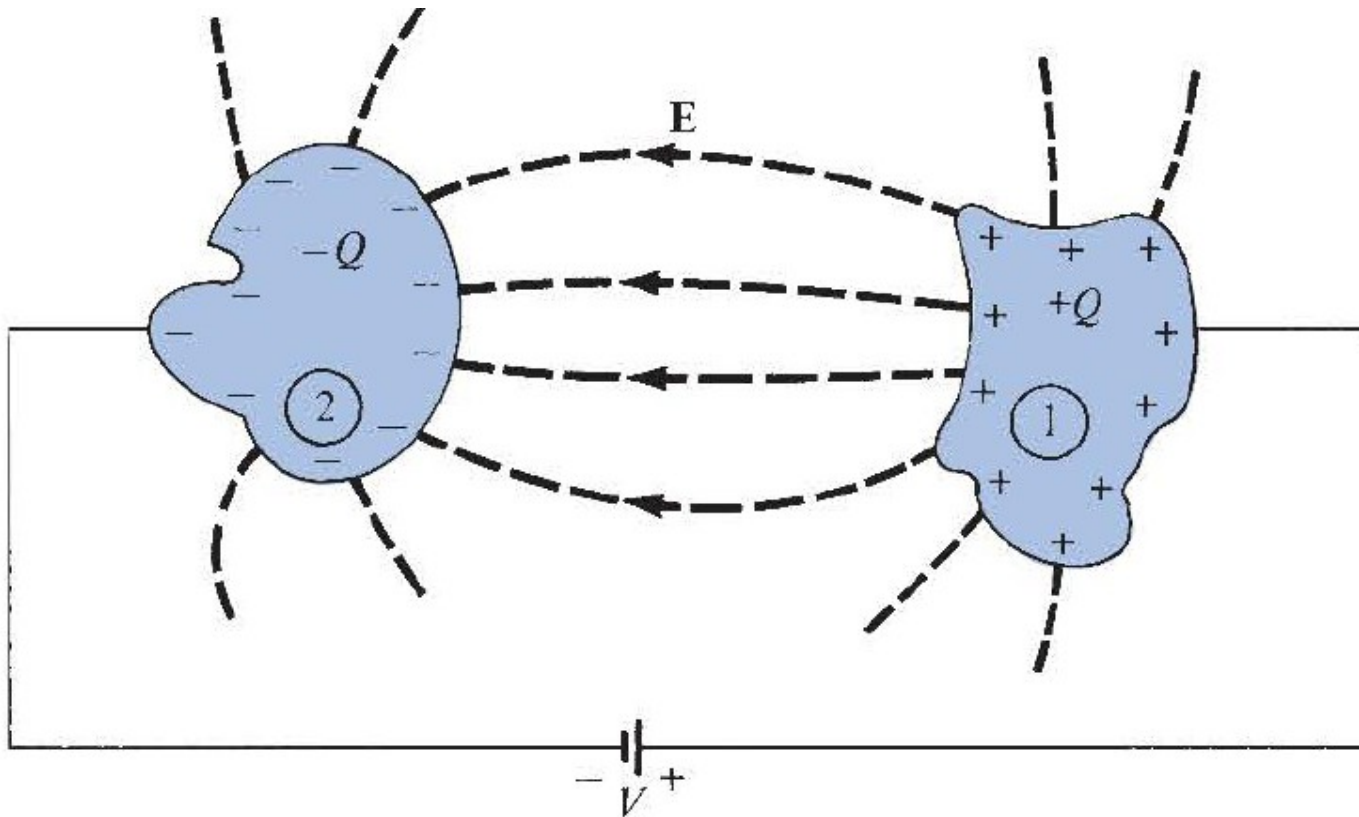
RESISTANCE & CAPACITANCE

- A capacitor must have two (or more) conductors carrying equal but opposite charges.
- This implies that all the flux lines leaving one conductor must necessarily terminate at the surface of the other conductor.
- The conductors are often referred as *plates* of the capacitor.
- The plates may be separated by free space or a dielectric.
- Consider the two-conductor capacitor as shown.
- The conductors are maintained at a potential



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



- Where E is the electric field existing between the conductors and conductor 1 is assumed to carry positive charge. field is always normal to the conducting surfaces.
- **Capacitance** of the capacitor is defined as the **ratio** of the magnitude of the charge on one of the plates **to** the potential difference between them
- The negative sign before Q has been dropped as we are interested in the absolute value of Q .

EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE

- The capacitance is a physical property of the capacitor and is measured in Farads.

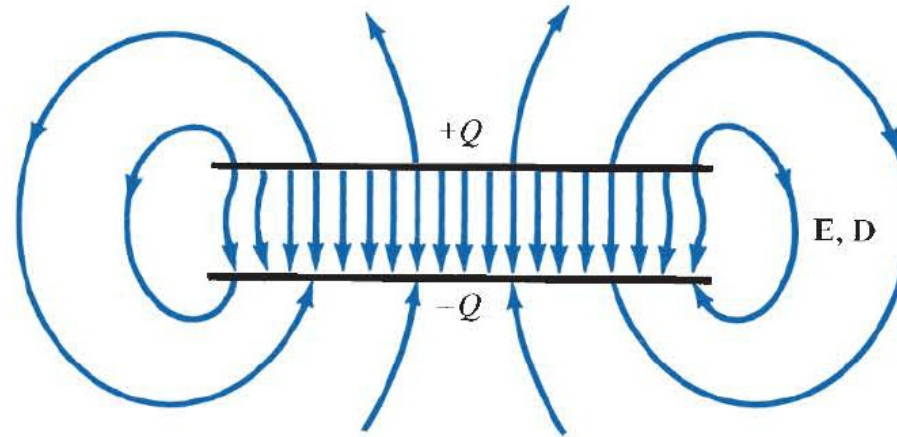
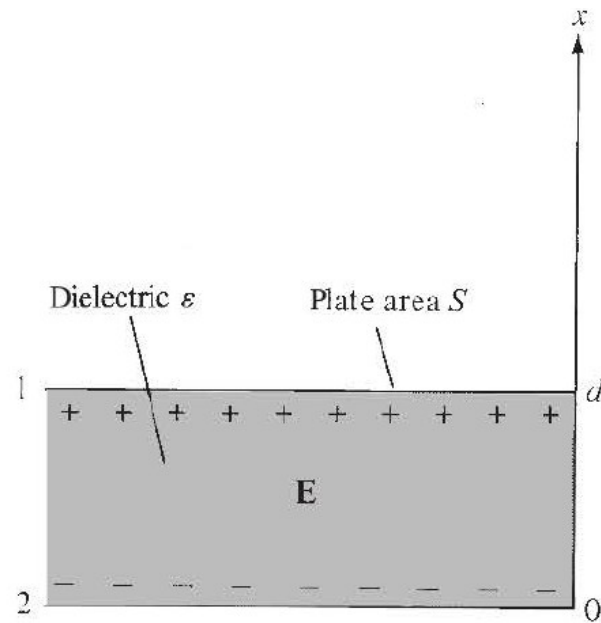
- Using the expression for , we can determine the capacitance of any given two-conductor capacitor by any one of the following methods;
- 1. Assuming and determining in terms of (using Gauss's Law)
- 2. Assuming and determining in terms of (using Laplace's Eqn.)
- Method 1 has the following steps;
- 1. Choose a suitable coordinate system
- 2. Let the two conducting plates carry charges and .
- 3. Determine using Coulomb's Law or Gauss's Law and find from The negative sign may be



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE

- 4. Finally, get from ____
- A. Parallel – Plate Capacitor: Consider a parallel-plate capacitor as shown.



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE

- Let each of the plates have a surface area and separated by a distance . Assume that plates 1 and 2, respectively, carry charges and uniformly distributed on them such that —
- An ideal parallel-plate capacitor is one in which the separation is very small compared with the dimensions of the plate.
- Assuming such an ideal case, the **fringing field** at the edge of the plates can be ignored so that the field between them is considered uniform
- If the space between the plates is filled with a homogenous dielectric with permittivity , we have:



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



- , recollect the field for a parallel - plate capacitor and direction of (-ve direction).
- Then ,
- Thus, for a parallel - plate capacitor .
- The formula stated above is also useful in measuring the dielectric constant of a given dielectric. By measuring the capacitance of a parallel - plate capacitor with the space filled with dielectric and capacitance with air between the plates, we get from .

EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE

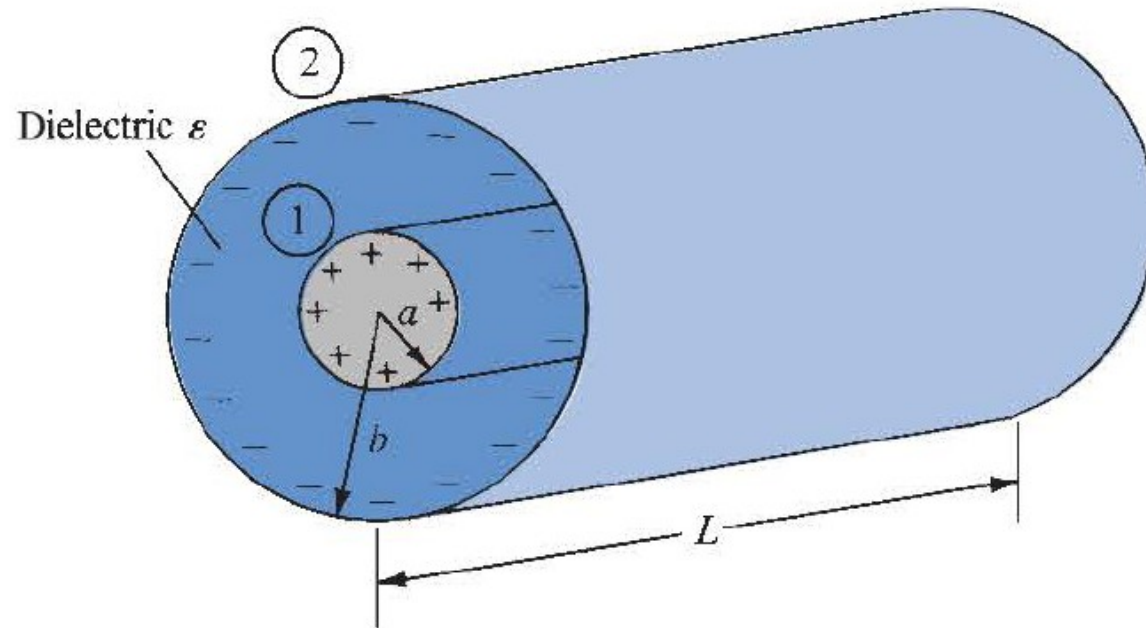
- The energy stored in parallel-plate capacitor is given by;

- .
- B. Coaxial Capacitor: A coaxial capacitor is essentially a coaxial cable or coaxial cylindrical capacitor.
- Consider a two coaxial conductor of length with inner radius and outer radius (r_2), as shown.
- The space between the conductors is filled with a homogeneous dielectric with permittivity ϵ .
- Assume the conductors carry Q and $-Q$, uniformly



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



- While applying Gauss's law to an arbitrary Gaussian Cylindrical surface of radius we get
- . This results in
- Neglecting fringe flux at the cylinder ends,
- . Note the limits !!! This takes care of
- .

EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE

- Then the capacitance of a coaxial cylinder is given as;

■

- C. Spherical Capacitor: It is a case of two concentric spherical conductors.

- Consider the sphere of radius r_1 and outer sphere of radius r_2 separated by a dielectric medium with permittivity ϵ as shown.

- Assume charges $+Q$ and $-Q$ on the inner and outer spheres, respectively.

- By applying Gauss's law to an arbitrary Gaussian

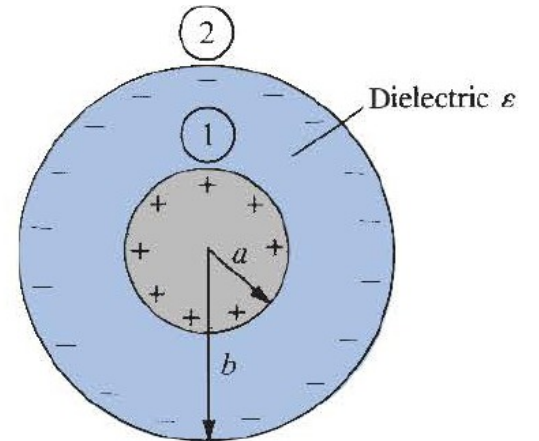


EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



- The following expressions;
- or .
- The potential difference between the conductors is;
- .
- Thus the capacitance of a spherical capacitor is;

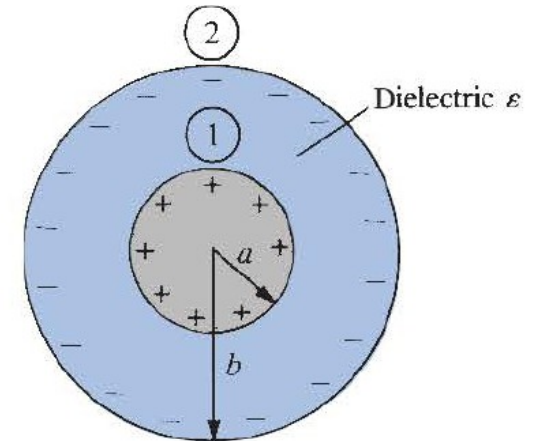


EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



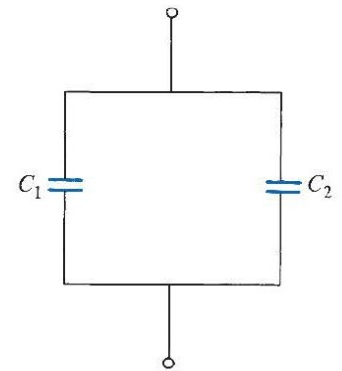
- By letting , which is capacitance of a spherical capacitor whose outer plate is infinitely large.
- Such is the case of a spherical conductor at a large distance from other conducting bodies – the *isolated sphere*.
- Even an irregularly shaped object of the same size as the sphere will have nearly the same capacitance.
- This is useful in estimating *stray capacitance* of an isolated body or equipment.



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE

- The total capacitance of two capacitors (having same charge) connected **in series** is , .
- When they are connected **in parallel** (having same voltage across the plates), total capacitance is .
- From the expressions;
- and , we also get
- which is the Relaxation Time of the medium separating the plates. **This relation is valid only if the medium is homogeneous.**



EMF Theory – Unit - 2

RESISTANCE & CAPACITANCE



- Assuming **homogeneous media**, the resistance of various capacitors dealt till now, can be summarised as;
- For Parallel - plate Capacitor, ,
- For Cylindrical Capacitor, ,
- For Spherical Capacitor, ,
- For isolated spherical conductor, ,

EMF Theory – Unit - 2

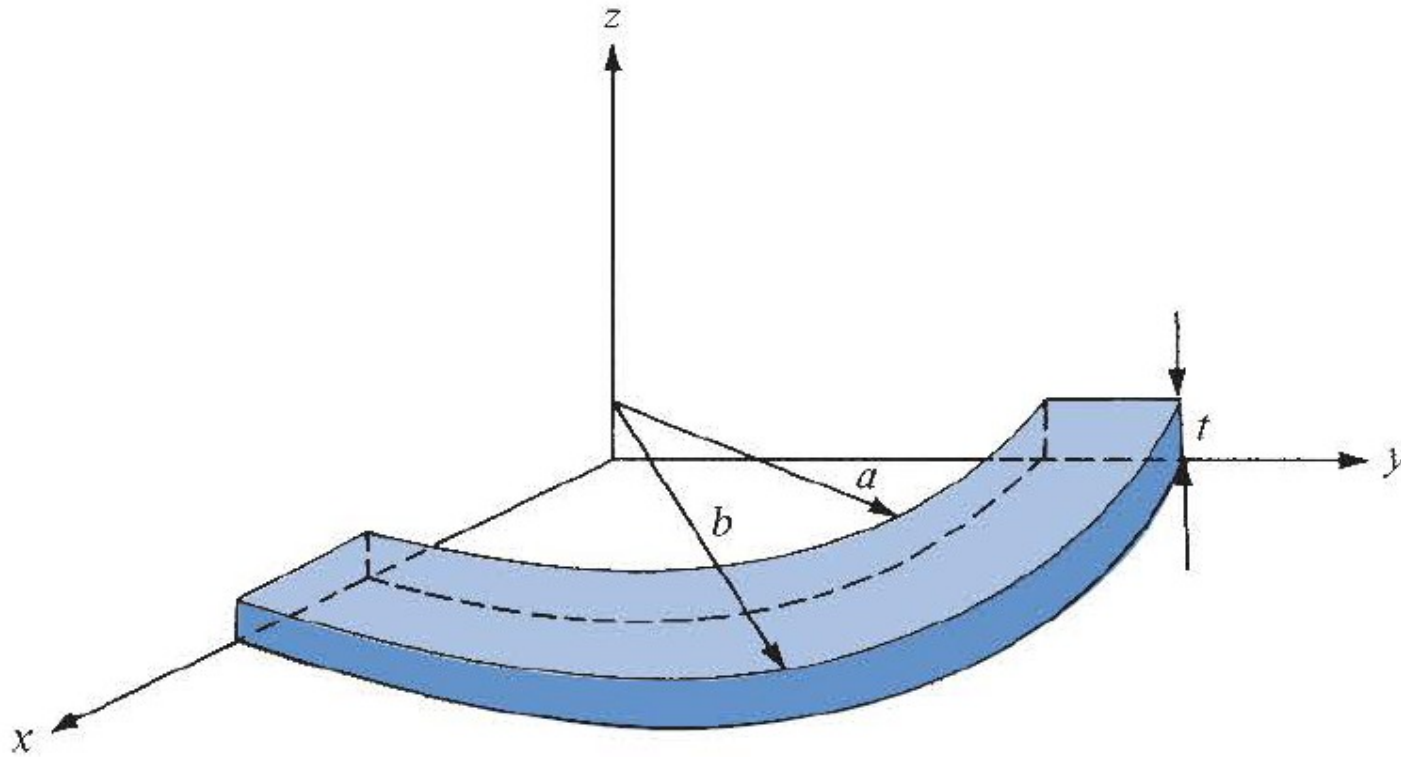
RESISTANCE & CAPACITANCE



- The resistance indicated above is the **leakage resistance** between the plates **and not** that of the capacitor plate !!!
- Therefore, the indicated in the equations is the **conductivity** of the **dielectric medium** separating the plates.
- **Numericals.**

EMF Theory – Unit - 2

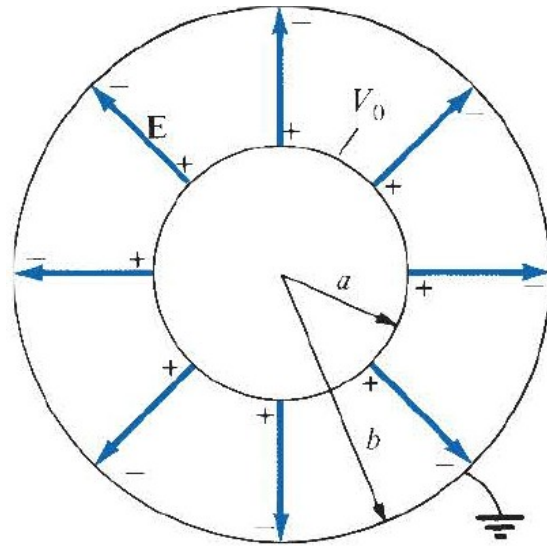
RESISTANCE & CAPACITANCE - NUMERICALS



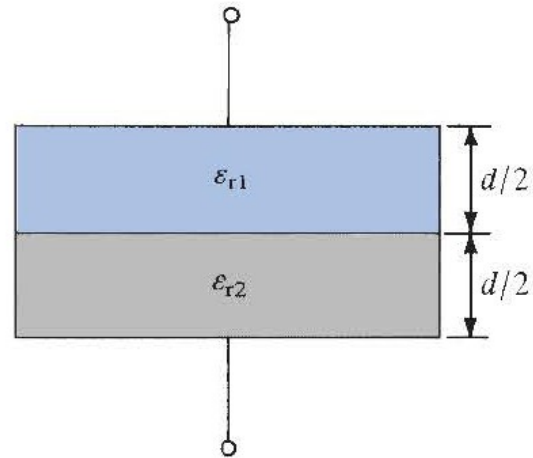
Ex.
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EMF Theory – Unit - 2

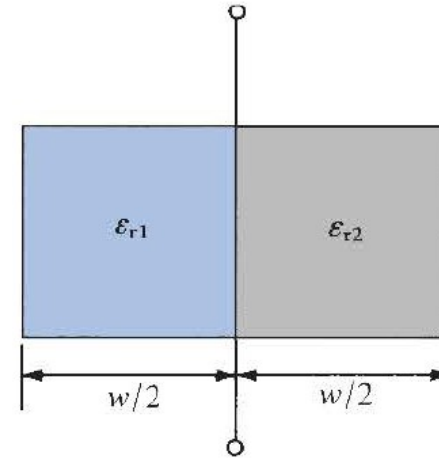
RESISTANCE & CAPACITANCE - NUMERICALS



Ex.
2



Ex.
3





THANK YOU

Dr. T. S. Chandar
Department of ECE
chandarts@pes.edu
+91 8197067914